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IET FQE - questions on iet

Radiography and imaging sciences (Kenya Medical Training College)

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KENYA MEDICAL TRAINING COLLEGE
NYERI CAMPUS
DEPARTMENT OF MEDICAL IMAGING SCIENCES

IET FQE 2019

SECTION B

Question 1

List the components of a tube filament circuit (5 marks).

Step-down transformer- Steps down the voltage, but steps up the current thus a low voltage and high current.

Variable resistor- Controls filament current and therefore filament temperature.

Ammeter- To give an indication of filament current which is directly related to Ma.

Focal-spot selector- A switch that allows current to flow through one filament wire at a time. When you select the large focal spot the filament current flows through the longer filament wire, and when you select small focus the filament current is switched to flow through the small filament wire.

mA selector- Measures current flow when rapid exposure time settings are employed.

Question 2

Describe the principle of image intensification **(5 marks)**

A visual image, which is a light pattern, is converted to an electron pattern and then to an electric current or voltage which will vary in proportion to the differing light intensities of the original (visual information becomes electrical information).

The varying electrical current is converted back to a light pattern which makes a visual impression on an observer similar to the original(electrical information becomes visual information).

Question 3

Justify the choice of a material used in the construction of a secondary radiation grid **(5 marks)**

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Radio-paque strips or slats usually made of lead or tungsten

Radiolucent spacing material of aluminium, paper or wood.

Protective covering, front and back, of thin aluminium or plastic.

Protective strips around the edges of the grid.

Question 4

State the changes that occur in the mains voltage **(5 marks)**

Slow changes over a period of time due to differences in demand on the supply at various priods of the day.

Rapid instantaneous changes due to current being drawn on the same line by equipment which takes high current for short intervals.

A fall in the mains voltage which occurs as soon as the x-ray exposure begins and is due to load current drawn by the x-ray set itself for the exposure

Question 5

Explain the importance of voltage compensation during X-ray exposure **(5 marks)**

To avoid change in the x ray tube kilovoltage due to alteration in the voltage supplied by autotransformer to primary winding of the high tension transformer.

To avoid change in the x ray tube current due to alteration in the voltage supplied by autotransformer and hence in heat of the filament. Small changes in filament heat can produce changes in the tube mA

To avoid change in the voltage supplied to transformers heating the filaments of rectifying valves in the high tension circuit. Fall in the voltage here leads to lowered temperature for the valve filaments and hence reduced electron emission.

It prevents change in voltages operating many components in the x-ray set

Question 6

Explain the frequency changes in the mains supply **(5 marks)**

Frequency changes occur due to differences in the speed of rotation of the alternators which occur slowly as the result of varying demands for electricity change. This must be so because the rate of repetition of the a.c wave pattern(that is the number of cycles per second or the frequency) depends on the speed of rotation of the alternators. So when the alternators rotate more slowly the mains frequency falls, and when they rotate more quickly the mains frequency rises.

Periods of heavy demand on generators causes frequency change. Change in frequency affects the static voltage stabilizers. This changes the voltageoutput. Increase in frequency causes increase in voltage and decrease in frequency leads to decrease in voltage. Changes in frequency

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causes irregularity in the voltage for the primary winding of the filament transformer hence changing the tube current.

Question 7

Describe the factors considered when selecting the material of choice for the X-ray tube target **(5marks)**

Melting point- The material should have a higher melting point to withstand higher temperatures.

Atomic number- It should have a fairly high atomic number thus more efficient at producing x-rays

Heat conductivity- It should be a fairly good conductor of heat to pass the heat reasonably quickly away from the small area where it is being produced.

Vaporization- It should not vapourize easily at high temperature.

Smoothness- To ensure x rays are produced from a source which is smooth.

Question 8

Describe the convection process of heat control in a rotating anode X-ray tube **(5 marks)**

Convection takes place in liquids and gases

First heat is removed from the target area by conduction into the copper block of anode.

The target also conveys a small part of its heat by radiation across the vacuum of x ray tube to the glass envelope.

Via anode and glass envelope, heat reaches oil in which the tube is immersed. Oil accepts the heat without great rise in temperature. Convection currents are set up in the oil and the heat is conveyed to the metal casing.

It is then conducted through the casing to its outer surface in contact with air in the room. The metal casing then loses heat by convection and radiation to the air surrounding the x ray tube.

Question 9

With a well labeled diagram, describe the high tension cable **(5 marks)**

Electrical conductors- These carry the supply to the x ray tube. In a cable to be used for the anode end of x ray tube, only one such inner conductor is necessary, but for a cable to be used at cathode end of dual focus tube three are necessary.

Insulating layer- Made of special rubber with very good insulating properties, put round the copper cores in a thick layer. The higher the voltage to which the cable is to be subjected the thicker this layer must be.

Flexible metallic sheath- Made of strands of metal plaited or braided together. It is connected to earth and constitutes the earthed metal screen.

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Plastic or cotton sheath- Designed to protect the flexible metal braiding from damage.

Question 10

Explain the two major importance of regular care and maintenance of imaging equipment **(5 marks)**

Regular care and maintainance ensures fewer equipment failures. This reduces costs. Where a failure occurs, local hospital staff may be able to repair without an expensive service call.

When an external service call is required, the service technician can arrive fully informed to deal with the situation, together with the appropriate equipment or parts. This will reduce the possibility, and expense, of repeated visits.

Repeat films will be kept to a minimum. Staff job satisfaction will increase.

Patients will receive less radiation and less inconvenience

Question 11

State the importance of incorporating safety devices in imaging equipment **(5 marks)**

To prevent wear and tear of equipment.

Provide protection to workers against exposure to radiation

Improve efficiency of the imaging equipment

Question 12

Describe the use of broad and fine focus spots in medical imaging **(5 marks)**

Broad focus spot- Are used for all radiographic examinations which require a large exposure dose (because the body part is thick) and a short exposure (because movement in subject or part of the subject is probable). This results in greater geometric unsharpness but greatly reduces the motional unsharpness, and the result is the sharper image than would be obtained from using a small focus.

Fine focus- Used in radiographic examinations in which the risk of movement is not very great. Generally used for examinations of bony parts of the body.

Question 13

Define magnetic resonance imaging **(5 marks)**

Its an non invasive scanning procedure that provides visualization of fluid, soft and bony structures without the use of radiation but magnet field and radio waves. After the patient has been positioned in the magnetic field a radio wave is sent into the body where they are transmitted and received by the transmit-receiver coil. The analog data is later converted to digital information for reconstruction.

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Question 14

Explain the terms; primary and secondary exposure switching **(5 marks)**

Primary switching involves making and breaking currents at voltages which have not been stepped up from the mains voltage to the kilovoltage necessary for operation of the x-ray tube. So primary switching involves the switching of high currents at mains voltage.

Secondary switching

Question 15

Describe collimation in Computed tomography **(5 marks)**

CT scan uses two collimators, pre-patient and post-patient collimators.

Post-patient collimator (at the detector) shape and limit the beam incident on the detectors to the sliced thickness. The pre-patient collimator(at the x-ray tube) limits the area of the patient that intercepts the useful beam and thereby determines the slice thickness and patient dose.

Collimators reduces patient dose and improves image quality

Question 16

State five advantages of an image orthicon camera tube **(5 marks)**

Can work satisfactorily at quite low levels of illumination.

It can reproduce rapid movement without blur.

Ideal/high sensitivity.

The signal to noise ratio is better.

It produces no lag

SECTION C

Question 1

Describe the principle and construction of the filament heating transformer **(20 marks)**

It's a step down transformer. For x-ray to be produced the filament must be heated so the electrons are emitted. It's convenient to heat the filament by passing current through it and not much power is needed to heat the extremely fine filaments of x-ray tubes to the necessary high temperature. The power is provided by a small step down transformer to the secondary winding of which the filament of the x-ray tube is directly connect.

Turns ratio

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The filament transformer provides a secondary voltage of 8-12 volts ,and this sents a heating current of 4-8 amperes through the filament of x-ray current.

The core

It's constructed similar to that of H.T.T

Its laminated and made of a special alloy so the losses in the core may be minimized.

Both the transformer windings of the filament transformer are put on one limb of the core,the secondary winding being wound on top of the primary one.

The transformer is required to provide only a small amount of power and so the core and the winding are much smaller in size than the same features of the H.T.T

WINDING

The primary winding has more turns in it than the secondary winding

Primary winding consist of many turns carrying small currents and viceversa in secondary winding

OIL IMMERSION

Its oil immersed in an earthed metal tank and oil functioning both to insulate and to cool a transformer.

Question 2

Explain the following terms as used in CT scan

i. Filtration **(5 marks)**

Radiation from x ray tubes consist of long and short wavelengths. Thus, filtration is the removal of long wavelength x rays because they don't play a role in CT image formation but instead contribute to patient dose.

ii. Collimation **(5 marks)**

Uses two collimators ,prepatien t(at the x-ray tube)and post patient(at the detector) which shape and limit the beam.The prepatient collimator determines the slice thickness. Collimator reduces patient dose and improves image quality

iii. Radiation detection **(5 marks)**

X-rays generated pass through the pre-patient collimator, patient, post-patient collimator and then the detector. In detectors, the x-rays are converted into

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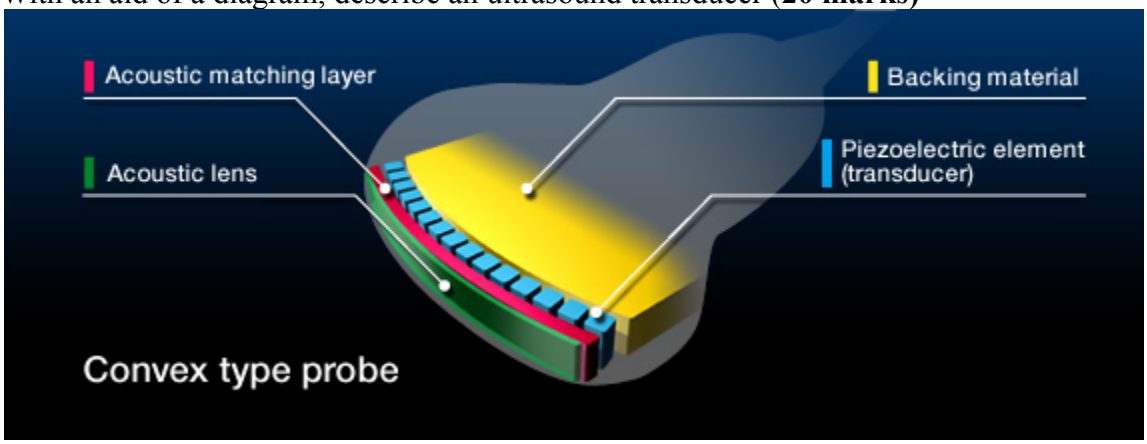
light photon energy. The light is converted into a digital image and sent electrically to the computer system.

iv. Detector array **(5 marks)**

Detectors are solid state composed of photodiodes coupled with scintillating crystal materials (cadmium tungstate or rare earth oxide ceramic crystals). Converts transmitted x - ray energy into light which is converted into electrical energy and then into a digital signal. Detector array affect patient dose and the efficiency of the CT unit**.**

Question 3

With an aid of a diagram, describe an ultrasound transducer (**20 marks)**



The ultrasound probe consist of a piezoelectric element,backing material,an acoustic matching layer and an acoustic lens.The functions of a piezoelectric trasducer include:

THE PIEZOELECTRIC ELEMENT(TRANSDUCER)

Is an essential part of the probe to generate ultrasonic waves.On both sides of piezoelectric element electrodes are affixed and a voltage is applied. The element then oscillates by repeatedly expanding and contracting,generating asound wave.When the element is externally applied with vibrations in turn, it generates a voltage.Piezoelectric ceramic is mostly used because of its high conversion efficiency.

THE BACKING MATERIAL

Its located behind the piezoelectric element to prevent excessive vibration.Reducing excessive vibration will cause the element to generate ultrasonic waves with ashorter pulse length,improving axial resolution.

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THE ACOUSTIC MATCHING LAYER

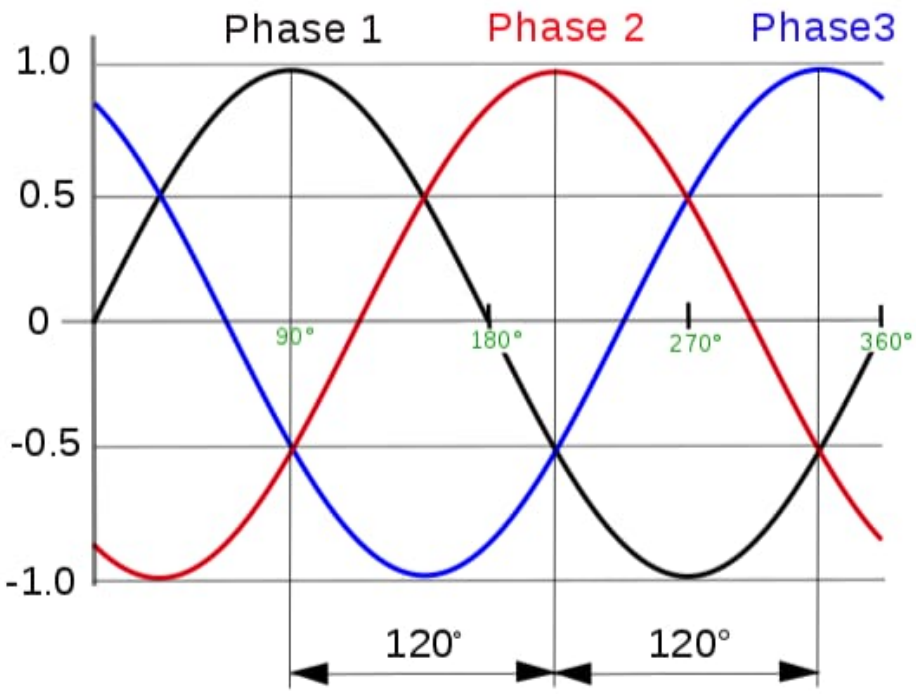
Ultrasonic waves transmitted from the piezoelectric element are reflected off a target because there is abig difference in acoustic impedance between the piezoelectric element and the object.To avoid this phenomenon,an intermediate material is inserted between the two so that ultrasonic waves can efficiently enter the object.This acoustic matching for less reflection of ultrasonic waves makes it possible to implement highly sensitive probes.Its designed to have adequate acoustic impeadance value using a combination of different resin material.

THE ACOUSTIC LENS

Is a gray coloured part looking like rubber attached to the tip of the probe.Ultrasonic waves transmitted from the probe would spread and travel like light.The acoustic lens prevents the ultrasonic waves from spreading and focuses them in the slice direction to improve the resolution.

Question 4

Describe the generation of a polyphase alternating current **(20 marks)**



One voltage cycle of a three-phase system

Its production of a number of sine waves simultaneously (only three).Its produced by three coils moving in relation to a magnetic field. The three coils are grouped round each other and displaced equally from each within 360 degrees. As the coil rotates together and each has

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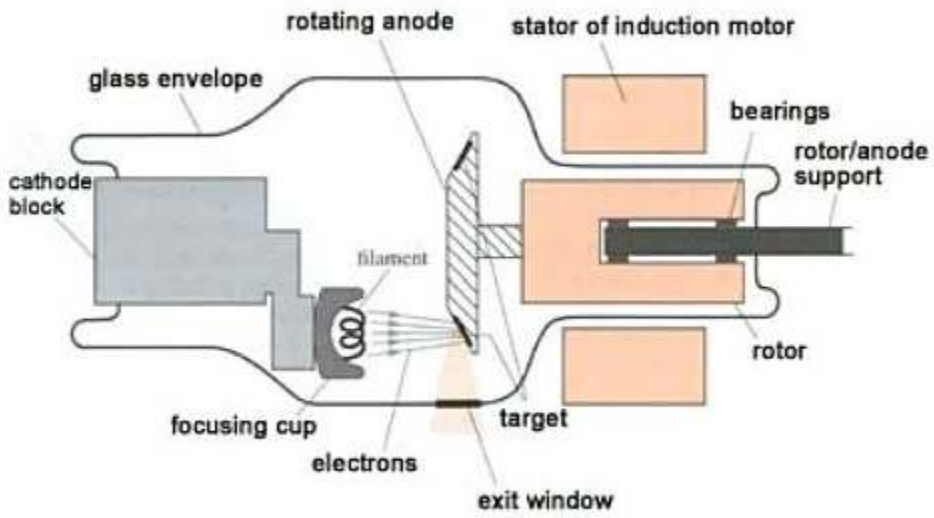
produced in it a separate alternating current sine wave e.m.f. All waves are of same pattern of all go up and down to peaks of the same height on vertical axis but they do not reach their peaks at the same instant of time hence outphase. When the coil is at zero position, the e.m.f is at zero value, when the coil is at 90 degrees the e.m.f is at positive maximum, when the coil is at180 degrees the e.m.f is at zero again,when the coil is at 270 degrees the e.m.f is at negative maximum and when the coil is at 360 degrees ,the emf is again at zero. So its possible to provide a generator of 3coils which can produce 3 separate supplies of alternating current .The voltage provided by 3 separate sources of electricity are out of phase with each other.

Question 5

Describe a rotating anode X-ray tube **(20 marks)**



THE CATHODE

Cathode consists of metal structure to support the filament which on operation will be heated so Cathode is connected to supply voltage. Without the supply there is no production of x-ray.

THE FILAMENT

The filament is heated to emit electrons. Electrons collect infront of the filament and forms cloud of electrons .The filament is made of tungsten because it has high melting point. The filament is enclosed so that the electrons are directed towards the anode. The filament is also off center to enable it to direct electrons to the beveled target.

ANODE

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It’s beveled at its edge to provide small area for x-ray production. It rotates during exposure; this is to allow at least each side of the anode to be heated. It’s supported by the stem .X-rays are produced in the anode.

GLASS ENVELOPE

It’s narrow at its edge. This is to allow for the fitting of the rotor. The glass envelope ensures a vacuum within the x-ray tube

STATOR AND ITS WINDING

Within it, there is electromagnetic induction which facilitates the rotation of the rotor.

ROTOR AND BALL BEARING

Rotor is made of copper because it’s a good conductor of heat. It conducts heat from the anode to the outside environment. It rotates on its ballbearing.The rotor is also painted black to facilitate heat loss from the anode.

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